PulsePatch Lite — Entity Relationship Diagram (ERD)

Local Pure C / FFI SQLite Engine (pulse_patch.db) • Schema Version: 2 (PRAGMA user_version = 2) • Active Integrity: Foreign Keys Enforced (PRAGMA foreign_keys = ON)

`Git Commit: 8b73e71 (master)`
Architecture Parity: 100% Verified (0 drift)

parsed in memory → stored as samples

High-Rate Telemetry — `samples`

PK	id	INTEGER
FK	session_id	TEXT
NN	t_ms	INTEGER
	bpm ?	REAL
NN	quality	INTEGER
NN	is_valid	INTEGER
	spo2 ? BLE RESV	REAL

1 : N — CASCADE

Core Sessions — `sessions`

PK	id	TEXT
NN	started_at	INTEGER
	ended_at ?	INTEGER
NN	type	TEXT
NN	quality_summary	TEXT
NN	app_version	TEXT
NN	source	TEXT

1 : 1 — CASCADE

Mathematical Summary — `session_stats`

PK FK	session_id	TEXT
NN	mean_hr	REAL
NN	std_hr	REAL
NN	min_hr	REAL
NN	max_hr	REAL
NN	start_hr	REAL
NN	end_hr	REAL
	recovery_drop ?	REAL
	pulse_interval_ms ?	REAL
	variability_est_ms ?	REAL
NN	valid_sample_count	INTEGER
NN	total_duration_seconds	INTEGER
	avg_spo2 ? BLE RESV	REAL

1 : 1 — CASCADE

On-Device AI Classification — `ai_results`

PK FK	session_id	TEXT
NN	model_version	TEXT
NN	label	TEXT
NN	features_json	TEXT
NN	created_at	INTEGER

STANDALONE SETTINGS

Daily Reminders — `reminders`

PK	id	INTEGER
NN	hour	INTEGER
NN	minute	INTEGER
NN	label	TEXT
NN	is_enabled	INTEGER
NN	days_of_week	TEXT

App Preferences — `app_settings`

PK	key	TEXT
NN	value	TEXT

ERD Legend & Notation Guide

Key Badges:

- PK — Primary Key (Unique entity identifier)
- FK — Foreign Key (Relational reference)
- NN — NOT NULL (Mandatory required column)
- ? — Nullable column (e.g. ended_at, recovery_drop)

Color Roles (App Visual Identity):

- sessions — Root measurement session entity
- samples — High-rate raw telemetry time-series
- session_stats — Derived mathematical features (11)
- ai_results — On-device tiny neural network inferences
- reminders / settings — Standalone user configurations

Referential Integrity & Cascade:

- All child tables enforce ON DELETE CASCADE with foreign keys ON.
- Deleting a session atomically removes samples, stats, and AI results.
- Standalone settings have zero foreign key relationships.

OUTSIDE THE DATABASE (Non-Relational Assets)

1. Physical Wearable Sensor Node (Embedded Hardware)

- Espressif ESP32-C3 RISC-V Microcontroller + Maxim MAX30102 High-Sensitivity Pulse Oximeter
- 3.3V Logic ONLY • SDA=GPIO4, SCL=GPIO5 • 100 Hz Optical Sampling (Red + IR) • IIR DC Filter

Transient physical state — analog voltage signals are sampled in registers, never stored directly in DB.

2. Over-The-Air BLE GATT Stream (Transient Ingestion)

- 12-Byte Little-Endian Heart Rate Notification Packet (Dispatched at 1 Hz, Protocol Version 0x01)

`[proto:1B][flags:1B][bpmX10:2B][sqi:1B][timeMs:4B][seq:2B][resv:1B]`

- Parsed in volatile memory by BlePacket.dart & gated through SignalQualityFilter (SQI ≥ 60)

Transient RF payload — once parsed and gated, valid fields are stored into `samples` table.

3. On-Device ML Model Weights Asset (Static JSON)

- Path: app/assets/ml/model_meta.json & ml/models/weights.json (Compiled Asset, NOT in DB)
- Architecture: Tiny MLP 5 → 16 → 8 → 1 (Exact Count: 216 weights + 25 biases = 241 parameters)

Pure Dart isolate execution (< 0.15 ms); inferences are recorded into `ai_results` table.

4. Offline Generated Biomonitoring Reports (Filesystem)

- Generated on-demand by PdfReportGenerator using PDF/Printing libraries into device app storage
- Aggregates `sessions`, `session_stats`, and `ai_results` into self-contained printable vector PDF

Stored as standalone binary .pdf files on disk, NOT as database BLOBs.

Engineering Data Boundary Summary

- Hardware & BLE streaming: ESP32-C3 sends 12-byte packets → Phone BLE layer parses in RAM.
- Data ingestion: Gated samples (SQI ≥ 60) are inserted row-by-row into `samples` table.
- Session closure: 11 summary features computed into `session_stats` within atomic transaction.
- ML execution: Dart Isolate reads `model_meta.json`, classifies pattern → stores in `ai_results`.
- Zero Cloud: 100% of telemetry, statistics, and configuration remain within local SQLite database.